

Surfaces and Area

Lecture 08

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Implicit Surfaces

Surfaces are often presented as level sets of a function, describe by an equation such as

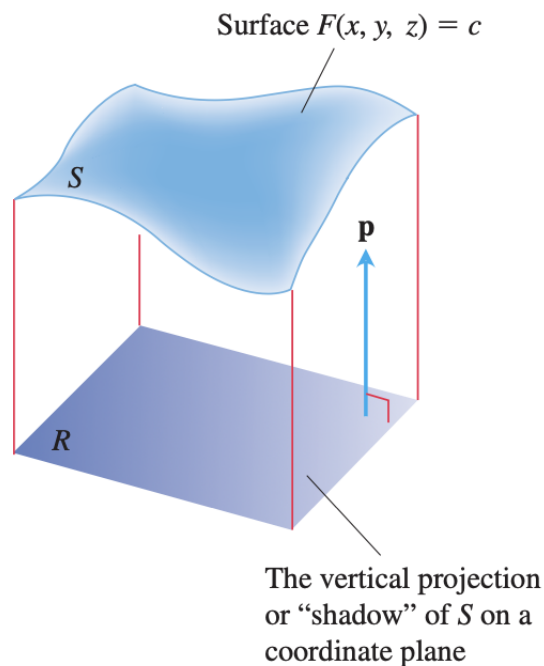
$$F(x, y, z) = c$$

for some constant c . Such a surface does not come with explicit parametrization, and is called an implicitly defined surface.

How to compute the surface area differential $d\sigma$ for the implicit surface?

Construction

The area of a surface S in space can be calculated by evaluating a related double integral over the vertical projection or “shadow” of S on a coordinate plane. The unit vector $\mathbf{p}$ is normal to the plane.



- Surface is smooth F is differentiable and $\vec{\nabla} F$ is non-zero and continuous on S
- $\vec{\nabla} F \cdot \mathbf{p} \neq 0$, so the surface never folds back over itself.

The area of the surfaces $F(x, y, z) = c$ over a closed and bounded plane region R is

$$\text{Surface Area} = \iint_R \frac{|\vec{\nabla} F|}{|\vec{\nabla} F \cdot \mathbf{p}|} dA$$

where $\mathbf{p} = \mathbf{i}, \mathbf{j}$, or $\mathbf{k}$ is normal to R and $\vec{\nabla} F \cdot \mathbf{p} \neq 0$.

Example

Find the area of the surface cut from the bottom of the paraboloid $x^2 + y^2 - z = 0$ by the plane $z = 4$.